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U-Net and Transformer-Based Remote Sensing Image Urban Road Recognition Method

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***Abstract*—In recent years, with the rapid development of remote sensing technology, urban remote sensing images have been widely used in urban planning, traffic management, disaster monitoring and other fields. However, the directions, colors, and shapes of urban roads are diverse. If a dataset that exhaustively enumerates all samples cannot be prepared, the recognition ability of the corresponding deep model will be very low. In order to solve the above problems, this paper proposes a U-Net and transformer-based remote sensing image urban road recognition method (UT-Road), a typical characteristic of this method is that it can recognize roads in cities based on "linear" recognition in the sequence through the transformer structure on the basis of the traditional semantic segmentation structure. In the experiment, we introduced multiple sets of remote sensing images for testing. The results showed that our method is significantly better than the traditional method in terms of accuracy and stability. It shows that UT-Road has huge application value in urban road management.**

Keywords- U-Net; Transformer; Urban Road Recognition; Deep learning; Remote Sensing

I. INTRODUCTION

Remote sensing images are a valuable source of information for a variety of applications, including urban planning, transportation management, and environmental monitoring [1]. Road recognition is an important task in remote sensing image analysis, which aims to extract road networks from remote sensing images. Accurately acquiring road information over a large area is of great value for urban management, infrastructure construction, and traffic management [2] [3]. Therefore, it is very necessary to carry out remote sensing image urban road recognition.

Traditional shallow learning methods, such as support vector machines (SVMs) and random forests (RFs), have been used for road recognition [4] [5]; However, these methods are limited by their inability to capture the complex and multi-scale features of road networks. Deep learning methods, such as convolutional neural networks (CNNs) and Fully Convolutional Networks (FCNs), have been shown to be effective for road recognition [6] [7]. However, these methods often require large amounts of training data and can be computationally expensive [8].

Traditional deep learning methods for road recognition typically follow a hierarchical feature extraction and classification paradigm. However, such methods struggle to handle the scale variation, complexity and diversity and data scarcity due to their limited ability to capture the global and contextual information of roads [9] [10] [11]. To address these limitations, this paper proposes a U-Net and transformer-based remote sensing image urban road recognition method (UT-Road), a typical characteristic of this method is that it can recognize roads in cities based on "linear" recognition in the sequence through the transformer structure on the basis of the traditional semantic segmentation structure. The experimental results demonstrate that our method significantly outperforms traditional methods in terms of classification accuracy.

II. METHODOLOGY

A. Overall process of UT-Road method

To solve the difficulties faced by current road recognition, the transformer structure is a typical solution, through transformer we can learn long-range dependencies in images (or image blocks), which can be helpful for road recognition. The overall process of U-Net and transformer-based remote sensing image urban road recognition method (UT-Road) can be seen as figure 1:

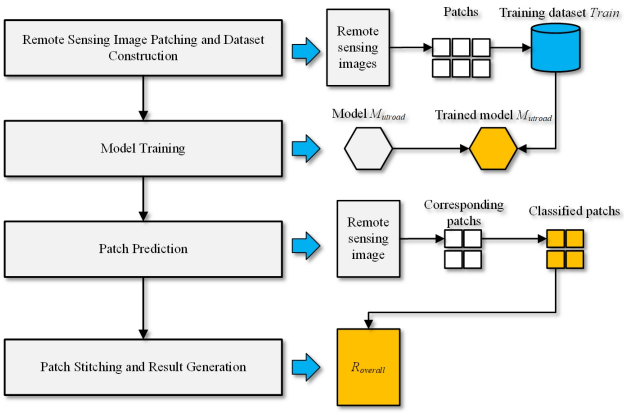


Figure 1. Overall process of UT-Road method.

As can be seen from figure1, the method is based on a deep learning model that combines the strengths of U-Net and transformer structures. UT-Road is a four-step method for urban road recognition from remote sensing images:

1) Remote Sensing Image Patching and Dataset Construction

The first step is to divide the input remote sensing image set I into fixed-size patches. Each patch is then manually labeled to obtain a training dataset. The size of the patches can be chosen based on the resolution of the remote sensing images and the desired accuracy of the road recognition results. The training dataset should be large enough to ensure that the model is able

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to learn the full range of possible variations in urban roads. Finally, we obtain the training dataset Train={t1, t2,…,tn}

2) Model Training

The training dataset is then used to train the UT-Road model. The trained model is denoted as Mutroad. The model can be trained using stochastic gradient descent (SGD):

$$wt+1 = wt - \alpha\nabla f(wt) \quad (1)$$

where, the wt represents the parameter of input image at the current iteration t. α represents the learning rate of Mutroad, which controls the step size of the update; and ∇f represents the gradient of the function evaluated at the current parameter vector. The gradient points in the direction of the steepest increase of the function. The Mutroad we obtain through this mode will have predictive ability.

3) Patch Prediction

For a given region, the region's image is divided into patches P={p1, p2, …, pn}. The model Mutroad is then used to predict the road probability for each patch. The prediction process can be parallelized to improve efficiency, and through this step we can obtain semantic segmentation result set based on patch Rpatch={r1, r2, …, rn};

4) Patch Stitching and Result Generation

To avoid obvious segmentation boundaries, the influence patch segment process in steps (2) and (3) can be overlapped with each other with an overlap degree of g. The overlapping areas are averaged during the stitching process. Then the result Roverall is then used to generate the final road recognition result.

Through the above four steps, a model Mutroad with the ability to identify urban roads can be obtained, and large-scale urban roads can be further automatically identified.

B. UT-Road model

In reality, the directions, colors (materials), and shapes of urban roads are diverse. It is difficult to construct a sample set that exhaustively enumerates all the spatial possibilities of these roads. This requires a deeper understanding of the characteristics of roads. Therefore, we propose the UT-Road model, the structure of which is shown in the figure 2.

As shown in Figure 2, the UT-Road model mainly consists of three components:

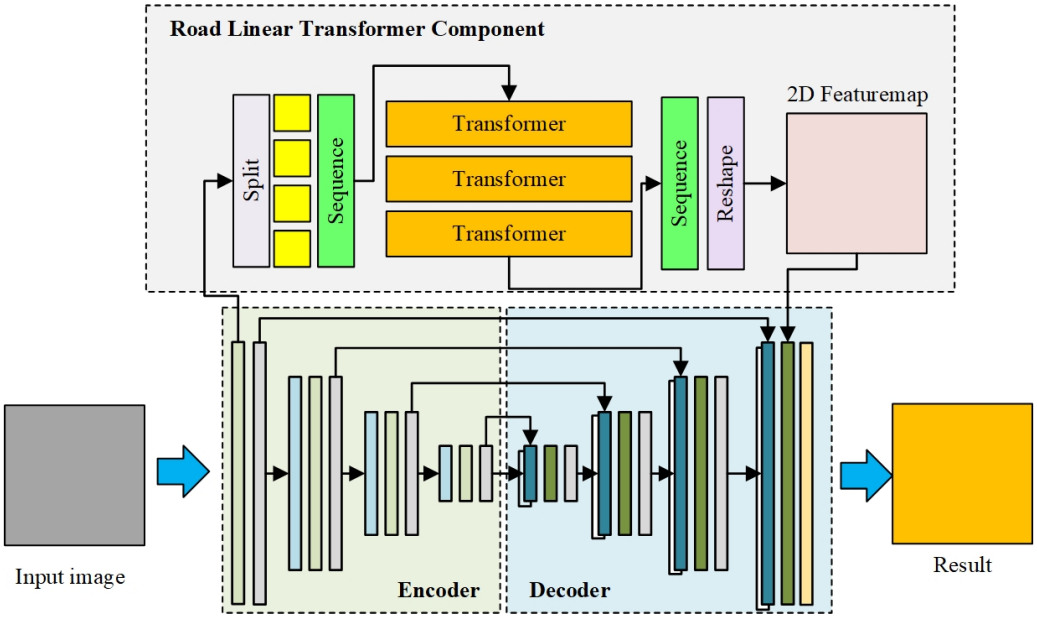


Figure 2. The structure of UT-Road model.

1) Encoder Component

The main function of this component is to input the remote sensing image block IPatch, extract spatial information from IPatch layer by layer and in groups, and obtain multi-scale expression of spatial information. At the same time, each group will transmit the obtained results to the corresponding decoder, so that the decoder can understand the spatial structure and scale when restoring spatial information. The encoder part of U-Net consists of a series of convolutional layers that are applied to the input image. Each convolutional layer is followed by a pooling layer, which reduces the size of the feature maps while preserving important information.

2) Road Linear Transformer Component

This component consists of three steps:

a) Input remote sensing image data, divide the image data into blocks, and use convolutional neural networks to extract n-dimensional attributes from the content of the blocks. This process can be expressed as:

$$Fpacth = f_CNN(BR(IRS)) \quad (2)$$

where Fpacth represents the n-dimensional attributes extracted from the block. The f_CNN represents the convolutional neural

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network function that extracts features; and BR(IRS) represents the function of dividing the remote sensing image data into blocks

b) Reshape the blocks into an n-dimensional sequence, and input the sequence to an m-head Transformer to extract key features from the sequence. This process can be expressed as:

$$Xseq = T_m(Reshape(Fpacth))) \quad (3)$$

where Xseq represents the n-dimensional sequence obtained by reshaping the blocks; T_m represents the m-head Transformer function that extracts key features.

c) Recombine the sequence output by the Transformer into a two-dimensional feature map, and further resize it to the group size of the Decoder module. Integrate the "linear" features into the decision-making and semantic segmentation results. This process can be expressed as:

$$Fout = G(Xseq) \quad (4)$$

where Fout represents the final two-dimensional feature map containing the integrated features; G represents a function that combines the Transformer output with potential information from the decoder.

3) Decoder Component

In the context of road recognition using remote sensing imagery, the Mutroad decoder plays a crucial role in reconstructing detailed road features. Unlike the encoder that extracts high-level semantic information about the scene, the decoder focuses on recovering the precise spatial details lost during the down-sampling process in the encoder. The decoder utilizes a series of up-sampling layers, often referred to as de-convolutional layers. These layers effectively increase the resolution of the feature maps obtained from the encoder, bringing them closer to the original image size.

With the above three structure, we can collect spatial information of remote sensing images layer by layer, similar to traditional semantic segmentation networks; more importantly, we divide the image data into blocks and use the Transformer structure to find the key connections between the blocks. In other words, we find the common patterns of roads behind the images. This pattern allows us to achieve good end-to-end road classification capabilities without enumerating all possible road feature combinations.

III. EXPERIMENTS

We implemented all algorithms using Python 3.9 and Pytorch on an Intel i9/GTX 3090 ti computer. To compare the effectiveness of the algorithms, we included the following algorithms for comparison:

1) MLP: Typical three-Layer shallow neural network model.

2) SegNET: Uses the SegNET network architecture to perform semantic segmentation on the input remote sensing image.

3) FCN: Uses convolutional neural networks to encode and identify data, and introduces FCN layers to predict semantic segmentation results.

4) U-Net: Uses a model that combines an encoder and decoder for identification.

5) UT-Road: The proposed method in this paper.

To test the algorithm, we constructed a dataset based on high-resolution remote sensing images. The dataset contains 200 1000×1000 urban test cases. 100 of these test cases were randomly selected as training data, and the other 100 were used as test data. For the test data, the average accuracy of all test images is used as the final result.

The overall classification accuracy comparison of the five methods is shown in Table 1.

Table 1. Overall Classification Accuracy Comparison of the Five Methods.

Method	Overall Accuracy (%)
MLP	52.31
SegNET	78.53
FCN	81.16
U-Net	83.22
UT-Road	90.62

As can be seen from table 1: Shallow models generally have lower classification accuracy. Deep models perform the better than traditional deep models. The proposed method significantly outperforms all other methods which overall accuracy=90.62%. To further compare the performance of all methods under the condition of limited sample size, the training data is further divided into subsets of 10, 20, 30, … 100 images, and the models are trained on each subset. The results are shown in Table 2.

Table 2. Overall Classification Accuracy Comparison of the Five Methods.

Training set size	Overall Accuracy (%)				
	MLP	SegNET	FCN	U-Net	UT-Road
10	43.11	52.98	51.82	63.78	67.37
20	40.85	52.59	56.35	62.76	68.93
30	42.11	57.66	60.67	71.86	73.04
40	47.25	64.25	61.78	73.32	75.74
50	40.57	63.35	63.61	76.13	77.86
60	43.68	66.49	68.1	77.31	80.09
70	44.16	68.75	73.92	79.88	82.28
80	47.34	72.25	74.55	81.55	84.24
90	50.24	74.38	76.76	82.26	89.19
100	52.31	78.53	81.16	83.22	90.62

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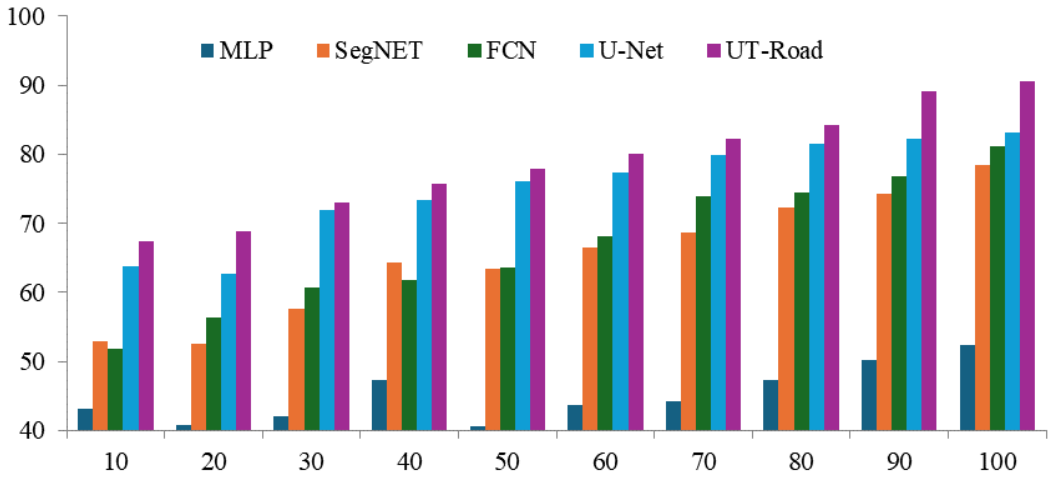


Figure 3. Accuracy Change Trend of Five methods.

The trend of accuracy change with the change of samples is shown in Figure 3.

As can be seen from Table 2 and Figure 3, the UT-Road method can achieve relatively high classification accuracy even when the sample size is small. With only 10 samples, the accuracy reaches 67.37%. With only 60 samples, the accuracy reaches 80.09%. This indicates that the proposed method can adapt very well to small sample situations. It also demonstrates that the use of Transformer structure can indeed extract road features from remote sensing images very well.

IV. Conclusion

In this paper, we proposed a novel method, UT-Road, for remote sensing image urban road recognition. The proposed method utilizes a U-Net architecture as the encoder and a Transformer-based decoder to extract road features from remote sensing images. The U-Net encoder captures the multi-scale contextual information of the image, while the Transformer decoder learns the sequence features of the image patches and utilizes them to generate the road probability map. We evaluated the proposed method on a variety of remote sensing image datasets and compared it with several state-of-the-art methods. The experimental results demonstrated that the proposed method achieves better classification accuracy than traditional deep learning methods and shallow models, especially when the training data size is limited.

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